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[4957]-1084

S.E. (Information Technology) (First Semester)

EXAMINATION, 2016

FUNDAMENTALS OF DATA STRUCTURES

(2012 PATTERN)

Time : Two Hours

Maximum Marks : 50

- N.B. :—**
- (i) Answer Q. No. 1 or Q. No. 2, Q. No. 3 or Q. No. 4, Q. No. 5 or Q. No. 6, Q. No. 7 or Q. No. 8.
 - (ii) Neat diagrams must be drawn wherever necessary.
 - (iii) Figures to the right indicate full marks.
 - (iv) Assume suitable data, if necessary.

1. (a) Explain various operators in C. [6]
- (b) Write a pseudo code to reverse a string without using library function. [4]
- (c) What is the purpose of enumeration type ? [2]

Or

2. (a) Describe the following declarations : [4]
- (i) `int*p[10]`
 - (ii) `int**q`
 - (iii) `int(*q)[5]`
 - (iv) `char s[50][50][50]`

P.T.O.

- (b) What is pointer ? Explain pointer to a function. [4]
- (c) Write a C program for copying the contents of one file into another. [4]
3. (a) What are different asymptotic notations ? [3]
- (b) Explain static and dynamic data structures with suitable examples. [3]
- (c) Show the output of each pass using selection sort to arrange following numbers in ascending order : [6]
- 15, 13, 100, 20, 72, 35, 65, 5, 99, 67.

Or

4. (a) Explain linear and binary search techniques with examples. [4]
- (b) Write pseudo C algorithm for quick sort. [6]
- (c) What is sort stability and efficiency ? [2]
5. (a) Write pseudo C algorithm for addition of two sparse matrices. [5]
- (b) Write pseudo C algorithm for simple transpose of sparse matrix. [4]
- (c) Represent the following polynomials using arrays : [4]
- (i) $7x^3 - 10xy + y^2 - 90$
- (ii) $x^4 + 47x + 50.$

Or

- 6.** (a) Write pseudo C algorithm for fast transpose of sparse matrix. [5]
- (b) Write a C code for stack as an ADT. [4]
- (c) What are applications of stack ? [4]
- 7.** (a) Write a C code for inserting a node at start and at end in SLL. [6]
- (b) List the applications of circular lists. [5]
- (c) What is the difference between malloc() and calloc(). [2]

Or

- 8.** (a) Explain GLL. Represent the following polynomial using GLL. [6]
- (L, (M, (N), (O, P)), Q), R, (S, T), (A, (B, C)).
- (b) Write a pseudo code to reverse SLL by changing link pointers. [7]